

VIBRATIONS OF TWO BEAMS ELASTICALLY COUPLED TOGETHER AT AN ARBITRARY ANGLE**

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ABSTRACT A general analytical method is developed for the vibrations of two beams coupled together at an arbitrary angle. The stiffness of a joint can take any value from zero to infinity to better model many real-world coupling conditions. Both flexural and longitudinal waves are included to account for the cross-coupling effects at the junctions. Each displacement component is here *invariantly* expressed, *regardless of the coupling or boundary conditions*, as a Fourier series supplemented by several closed-form functions to ensure the uniform convergence of the series expansions. Examples are presented to compare the current solution with finite element and experimental results.

KEY WORDS coupled beams, beams, free vibrations

I. INTRODUCTION

Determining the vibrations of beam-like structural members and power flow between them is of vital importance in the study of structure-borne sound propagation in many real-world structures. Extensive researches have resulted in a number of analytical, numerical, asymptotical and empirical methods that are applicable to structures and junctions of varying degree of complications in different (low, medium, and high) frequency regions.

The rapid progresses in computer technologies and computational methods have now made numerical methods, such as finite element methods, a standard tool for the dynamic analysis of complex structures. Although the immediate results from an FEA model are the displacements at nodal points, the power flows between different components, if desired, can be simply calculated by post-processing the nodal results^[1–4]. It is widely believed that the capability of the FEA methods is only limited by the computing resources. However, there are actually some other intrinsic reasons that make the FEA methods ineffective even useless at high frequencies, regardless of the computer resource consideration^[5]. While the idea of dividing a complex structure into a number of elements is theoretically necessary to faithfully represent its geometry, it is no longer suitable for a high frequency analysis because the extravagant inclusion of the insignificant details tends to make an FEA model not only numerically unmanageable, but also physically vulnerable to the input errors inherent to model parameters and manufacturing processes.

Many analytical methods have been developed in the literature to extend the prediction capabilities to higher frequency ranges. Among them, the Dynamic Stiffness Methods (DSM) are often used to

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calculate the power flows between beams/plates or in frameworks^[6–8]. In the dynamic stiffness methods, the displacements at both ends of a beam are determined in terms of the complex amplitudes for each wave component from which one can derive the relationship between the general force and displacement vectors. Because the physical meaning of each wave component is well understood as the propagating and evanescent non-propagating waves, the wave transmission coefficients are easily determined from the complex wave amplitudes.

Spectral Element Method (SEM) is another powerful technique for the vibration and energy flow analysis of simple or complex beam systems^[9–11]. In SEM each uniform beam or rod can be considered as a super-element on which the flexural, longitudinal and torsional waves will be expressed in terms of exact beam solutions. As a result, the coupling of various waves at a junction can be easily dealt with by following a FE-like assembling procedure. In comparison with the FE methods, an SEM model is much smaller with respect to the total number of degrees of freedom in the system, and more suitable for the vibration and power flow analyses at relatively high frequencies.

Analytical techniques based on receptance theory can also be used to study the energy flows between structures like rods, beams and plates^[12–14]. In the receptance methods, the relationship between the coupling forces and the displacements at the ends of a beam is derived by using Green's functions for the uncoupled beam under boundary conditions compatible with the entire system. The Green's function can be expressed either in a closed-form or as the expansion of the modes for the beam with appropriate boundary conditions. The receptance methods has also been employed to model more complicated junctions involving a number of connecting beams and the interactions of the flexural, longitudinal, and torsional waves^[15,16]. The advantage of the receptance theory methods is that each beam is considered as a fundamental element and an FEA-like assembling process can be followed to allow it to be used for composite structures.

Although a suite of techniques are available for the dynamic analysis of coupled beams or beam systems, there are various technical and practical issues and limits associated with each of them. For instance, the applications of SEM and DSM are primarily limited to beam frameworks because there is no exact form of general solutions for other types of structural components such as plates or shells. In addition, the beams involved have to possess certain degree of uniformity in material and geometrical properties; for example, a beam cannot have a varying mass density or cross-sectional area. In dealing with a beam structure, the system solution is often expressed in terms of the modal properties for each individual beam by assuming the coupling end(s) is free. The modal properties obtained may also be alternatively used to determine the transfer functions, Green functions or receptance functions between the responses and the reaction forces (including moments) at the junctions in the actual system environment. While this solution procedure may be reasonably good for calculating the modal properties or the vibrational response of the composite structure to an external load, the basic fact still cannot be changed; that is, the modal properties for each beam are actually determined by freeing the “coupling” end(s). As a consequence, the displacement functions constructed using these component modes will not be able to accurately represent the reaction forces/moments (which depend upon the second and third derivatives of the displacement functions) at the coupling ends. This problem is expected to become more remarkable in a power flow analysis because the reaction forces and moments will have to be calculated explicitly at the junctions. Although for an elastic joint the coupling forces and moments can be calculated from the relative translational and rotational displacements, respectively; the calculations will undoubtedly break down when the coupling stiffnesses become very high. As a matter of fact, the use of free-free beam functions was found to be particularly difficult for the intermediate coupling strength^[17]. Other well-recognized problems and concerns involve the numerical break-down of the standard beam functions approximately after the first dozen modes, the slow convergence of the modal expansions when the boundary/coupling conditions cannot be assumed correctly, and so on.

In view of the aforementioned issues and concerns, the development of a robust and sophisticated method which is capable of simplifying solution algorithms, reducing model input data, and universally dealing with various coupling and boundary conditions will be of great interest to both researchers and engineers. In a previous paper, a Fourier series method was derived for predicting the vibrations of two co-linear coupled beams and power flows between them^[18]. In this study, this solution method is extended to two beams elastically coupled together at an arbitrary angle. Accordingly, the flexural

and longitudinal displacements will have to be simultaneously determined to account for the effects of the cross-couplings at the junctions.

II. VIBRATIONS OF TWO ARBITRARILY COUPLED BEAMS

Figure 1 shows two beams that are connected together at an arbitrary angle θ . A pair of linear and rotational springs is used here to represent a general coupling between these two beams. The finite stiffness values for the springs will represent a flexible joint between the beams. The rigid connection is essentially obtained by setting the spring stiffnesses substantially larger than the bending rigidities of the involved beams. Similarly, the beams are assumed to be elastically restrained at the ends. Thus, any classical homogeneous boundary conditions (such as simply supported, clamped, or guided boundary conditions) can be considered as a special case when the stiffnesses of these elastic restraints become extremely large or small.

The differential equation for the transverse vibration of a beam is

$$D_i \frac{d^4 w_i(x)}{dx^4} - \rho_i S_i \omega^2 w_i(x) = f_{wi}(x) \quad (i = A, B) \quad (1)$$

where w_i , D_i , ρ_i and S_i are the flexural displacement, the bending rigidity, the mass density and the cross-sectional area of beam i , respectively; f_{wi} is the distributed load acting on beam i and ω is the frequency in radian form.

The differential equation for the longitudinal vibration is

$$E_i S_i \frac{d^2 u_i(x)}{dx^2} + \rho_i S_i \omega^2 u_i(x) = f_{ui}(x) \quad (i = A, B) \quad (2)$$

where u_i and E_i are the longitudinal displacement and Young's modulus for beam i , respectively; and f_{ui} is the distributed load in the axial direction.

It is evident from Fig.1 that, for an angled connection, the transverse vibration of beam A will be directly coupled with both the transverse displacement and axial displacement in beam B , and *vice versa*.

Following the previously developed technique^[17,18], the transverse and axial displacements on beam i will be sought as

$$w_i(x) = \sum_{m=0}^{\infty} C_{wi}^m \cos(\lambda_{im}x) + \phi_{wi}(x) \quad (3)$$

and

$$u_i(x) = \sum_{m=0}^{\infty} C_{ui}^m \cos(\lambda_{im}x) + \phi_{ui}(x), \quad 0 \leq x \leq L_i \quad \left(\lambda_{im} = \frac{m\pi}{L_i} \right) \quad (4)$$

where L_i is the length of beam i .

It is seen that the current Fourier series solutions are mathematically as simple as, if not simpler than, the traditional ones expressed in the form of modal expansions; for example, the cosine functions are obviously simpler than the beam functions for a free-free beam, and the auxiliary functions are comparable to the rigid-body modes which have to be included in the modal expansions in dealing with coupled-beam systems. Unlike in solving single-beam problems, the beam functions are used as a complete set to represent the displacement functions for each beam in the system because the governing equations and boundary/coupling conditions will have to be satisfied explicitly in solving the system. In this regard, the Fourier series is far more advantageous since it shows an excellent numerical stability and its convergence characteristic is proved rigorously, which is not true for the standard beam functions.

In Eq.(3), the auxiliary functions $\phi_{wi}(x)$ are introduced for the purpose of overcoming the discontinuity of the original displacement functions and their relevant derivatives appearing at the ends of each beam when they are periodically extended onto the entire x -axis by a Fourier series expansion. They are specifically required, *regardless of boundary/coupling conditions*, to satisfy the following equations:

$$\phi_{wi}'''(0) = w_i'''(0) = \alpha_{wi0} \quad (5)$$

$$\phi_{wi}'''(L_i) = w_i'''(L_i) = \alpha_{wi1} \quad (6)$$

$$\phi_{wi}'(0) = w_i'(0) = \beta_{wi0} \quad (7)$$

and

$$\phi'_{wi}(L_i) = w'_i(L_i) = \beta_{wi1} \quad (8)$$

The functions $\phi_{ui}(x)$ needs to satisfy only the two end conditions as given in Eqs.(7) and (8). The advantages of using such an auxiliary function were discussed before^[19,20], and will not be further elaborated here.

It can be proved mathematically that the series given in Eq.(3) (or Eq.(4)) is able to expand and uniformly converge to *any* function $f(x) \in C^3$ (or $g(x) \in C^1$) for $\forall x \in [0, L]$. More explicitly, the exact transverse (or longitudinal) displacement function for a beam, whether its closed-form exists or not, can always be expressed as the series expansion given in Eq.(3) (or Eq.(4)). It must be pointed out this statement does not depend upon what types of boundary or coupling conditions are involved or whether the beam is a single structure or a part of a complex structural system. In addition, the series given in Eq.(3) (or Eq.(2)) can be simply differentiated term-by-term to obtain the uniformly convergent series expansions for the first four (or two) derivatives of the displacement function. Mathematically, an *exact* beam solution refers to a *particular* function $w(x) \in C^3$ for $\forall x \in [0, L]$ which satisfies the governing differential equation(s) *exactly at every* field point and the boundary conditions *exactly at every* boundary point. Thus, the remaining task is to simply find such sets of expansion coefficients that the displacement functions in the form of series expansions will satisfy the governing equations and the boundary/coupling conditions in either an exact or an approximate manner. In the following, an exact approach will be adopted by letting the series exactly satisfy all governing equations and corresponding boundary and coupling conditions.

The boundary and coupling conditions can be written as

$$k_0 w_A(0) = -D_A w_A'''(0) \quad (9)$$

$$K_0 w_A'(0) = D_A w_A''(0) \quad (10)$$

$$k_1 w_B(L_B) = D_B w_B'''(L_B) \quad (11)$$

$$K_1 w_B'(L_B) = -D_B w_B''(L_B) \quad (12)$$

$$D_A w_A'''(L_A) = k[w_A(L_A) - w_B(0)] \quad (13)$$

$$D_A w_A''(L_A) = -K[w_A'(L_A) - w_B'(0) \cos \theta + u_B(0) \sin \theta] \quad (14)$$

$$D_B w_B'''(0) = -k[w_B(0) - w_A(L_A)] \quad (15)$$

$$D_B w_B''(0) = K \cos^2 \theta [w_B'(0) - w_A'(L_A) \cos \theta] + K_a \sin^2 \theta [w_B'(0) - u_A(L_A) \sin \theta] \quad (16)$$

$$E_A S_A u_A'(0) = k_{a0} u_A(0) \quad (17)$$

$$E_B S_B u_B'(L_B) = -k_{a1} u_B(L_B) \quad (18)$$

$$E_A S_A u_A'(L_A) = -k_a [u_A(L_A) - u_B(0) \cos \theta + w_B(0) \sin \theta] \quad (19)$$

$$E_B S_B u_B'(0) = k_a \cos^2 \theta [u_B(0) - u_A(L_A) \cos \theta] + k \sin^2 \theta [u_B(0) - w_A(L_A) \sin \theta] \quad (20)$$

where k , K and k_a are the stiffnesses of the coupling springs (refer to Fig.1) and those with subscripts 0 and 1 denote the stiffnesses of the springs at the left and right end of the system, respectively.

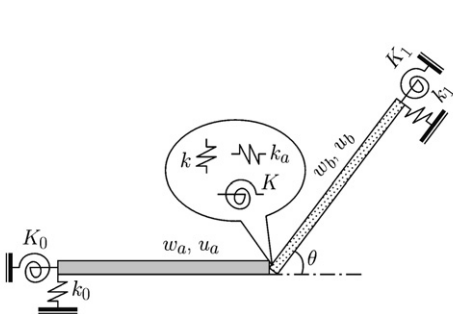


Fig. 1. Two beams connected at an arbitrary angle.

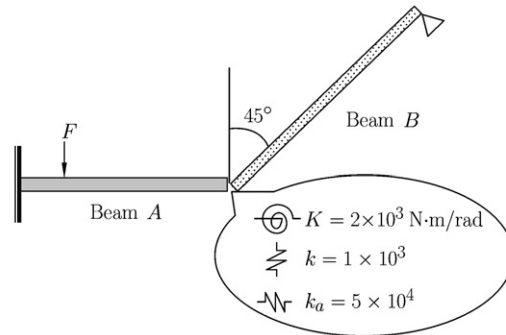


Fig. 2. Two beams elastically connected together at an angle of 45 degrees.

Theoretically, the auxiliary functions can be any continuous function in a closed form. Specifically, the auxiliary function $\phi_{wi}(x)$ will be here chosen as a simple polynomial

$$\phi_{wi} = \zeta_{wi}(x)^T \bar{\alpha}_{wi} \quad (21)$$

where

$$\bar{\alpha}_{wi} = \{\alpha_{wi0}, \alpha_{wi1}, \beta_{wi0}, \beta_{wi1}\}^T \quad (22)$$

and

$$\zeta_{wi}(x) = \begin{Bmatrix} -(15x^4 - 60L_i x^3 + 60L_i^2 x^2 - 8L_i^4)/(360L_i) \\ (15x^4 - 30L_i^2 x^2 + 7L_i^4)/(360L_i) \\ (6L_i x - 2L_i^2 - 3x^2)/(6L_i) \\ (3x^2 - L_i^2)/(6L_i) \end{Bmatrix} \quad (23)$$

The auxiliary function $\phi_{ui}(x)$ for the axial displacement can be obtained by only taking the last two elements in Eqs.(22) and (23), i.e.,

$$\zeta_{ui}(x) = \begin{Bmatrix} (6L_i x - 2L_i^2 - 3x^2)/(6L_i) \\ (3x^2 - L_i^2)/(6L_i) \end{Bmatrix} \quad (24)$$

It should be pointed out that although the same symbol x is used in Eqs.(23) and (24), it actually represents two different (local) coordinate systems with their origins located at the left ends of the corresponding beams, respectively. The use of a local coordinate on each beam is merely for the sake of the convenience of derivations and the consistency of the final expressions.

By substituting Eqs.(3)-(8) and (21)-(24) into the boundary and coupling conditions, Eqs.(9)-(20), one will have

$$k_0 \left(\sum_{m=0}^{\infty} C_{wA}^m + \frac{8L_A^3 \alpha_{wA0}}{360} + \frac{7L_A^3 \alpha_{wA1}}{360} - \frac{\beta_{wA0} L_A}{3} - \frac{\beta_{wA1} L_A}{6} \right) = -D_A \alpha_{wA0} \quad (25)$$

$$K_0 \beta_{wA0} = D_A \left(- \sum_{m=1}^{\infty} \lambda_{Am}^2 C_{wA}^m - \frac{\alpha_{wA0} L_A}{3} - \frac{\alpha_{wA1} L_A}{6} + \frac{\beta_{wA1}}{L_A} - \frac{\beta_{wA0}}{L_A} \right) \quad (26)$$

$$k_1 \left[\sum_{m=0}^{\infty} (-1)^m C_{wB}^m - \frac{7L_B^3 \alpha_{wB0}}{360} - \frac{8L_B^3 \alpha_{wB1}}{360} + \frac{\beta_{wB0} L_B}{6} + \frac{\beta_{wB1} L_B}{3} \right] = D_B \alpha_{wB1} \quad (27)$$

$$K_1 \beta_{wB1} = -D_B \left[\sum_{m=1}^{\infty} (-1)^{m+1} \lambda_{Bm}^2 C_{wB}^m + \frac{\alpha_{wB0} L_B}{6} + \frac{\alpha_{wB1} L_B}{3} + \frac{\beta_{wB1}}{L_B} - \frac{\beta_{wB0}}{L_B} \right] \quad (28)$$

$$\begin{aligned} k \left[\sum_{m=0}^{\infty} (-1)^m C_{wA}^m - \frac{7L_A^3 \alpha_{wA0}}{360} - \frac{8L_A^3 \alpha_{wA1}}{360} + \frac{\beta_{wA0} L_A}{6} + \frac{\beta_{wA1} L_A}{3} \right] = \\ D_A \alpha_{A1} + k \cos \theta \left(\sum_{m=0}^{\infty} C_{wB}^m + \frac{8L_B^3 \alpha_{wB0}}{360} + \frac{7L_B^3 \alpha_{wB1}}{360} - \frac{\beta_{wB0} L_B}{3} - \frac{\beta_{wB1} L_B}{6} \right) \\ + k \sin \theta \left(\sum_{m=0}^{\infty} C_{uB}^m - \frac{\beta_{wB0} L_B}{3} - \frac{\beta_{wB1} L_B}{6} \right) \end{aligned} \quad (29)$$

$$K \beta_{wB0} = K \beta_{wA1} + D_B \left(- \sum_{m=1}^{\infty} \lambda_{Bm}^2 C_{wB}^m - \frac{\alpha_{wB0} L_B}{3} - \frac{\alpha_{wB1} L_B}{6} + \frac{\beta_{wB1}}{L_B} - \frac{\beta_{wB0}}{L_B} \right) \quad (30)$$

$$\begin{aligned} D_B \alpha_{wB0} + (k \cos^2 \theta + k_a \sin^2 \theta) \left(\sum_{m=0}^{\infty} C_{wB}^m + \frac{8L_B^3 \alpha_{wB0}}{360} + \frac{7L_B^3 \alpha_{wB1}}{360} - \frac{\beta_{wB0} L_B}{3} - \frac{\beta_{wB1} L_B}{6} \right) = \\ k \cos^3 \theta \left[\sum_{m=0}^{\infty} (-1)^m C_{wA}^m - \frac{7L_A^3 \alpha_{wA0}}{360} - \frac{8L_A^3 \alpha_{wA1}}{360} + \frac{\beta_{wA0} L_A}{6} + \frac{\beta_{wA1} L_A}{3} \right] \\ - k_a \sin^3 \theta \left[\sum_{m=0}^{\infty} (-1)^m C_{uA}^m + \frac{\beta_{uA0} L_A}{6} + \frac{\beta_{uA1} L_A}{3} \right] \end{aligned} \quad (31)$$

$$K\beta_{wA1} = K\beta_{wB0} - D_A \left[\sum_{m=1}^{\infty} (-1)^{m+1} \lambda_{Am}^2 C_{wA}^m + \frac{\alpha_{wA0} L_A}{6} + \frac{\alpha_{wA1} L_A}{3} + \frac{\beta_{wA1}}{L_A} - \frac{\beta_{wA0}}{L_A} \right] \quad (32)$$

$$k_{a0} \left(\sum_{m=0}^{\infty} C_{uA}^m - \frac{\beta_{uA0} L_A}{3} - \frac{\beta_{u1} L_A}{6} \right) = E_A S_A \beta_{uA0} \quad (33)$$

$$k_{a1} \left[\sum_{m=0}^{\infty} (-1)^m C_{uB}^m + \frac{\beta_{uB0} L_B}{6} + \frac{\beta_{uB1} L_B}{3} \right] = -E_B S_B \beta_{uB1} \quad (34)$$

$$\begin{aligned} k_a \left[\sum_{m=0}^{\infty} (-1)^m C_{uA}^m + \frac{\beta_{uA0} L_A}{6} + \frac{\beta_{uA1} L_A}{3} \right] + E_A S_A \beta_{uA1} = \\ -k_a \sin \theta \left(\sum_{m=0}^{\infty} C_{wB}^m + \frac{8L_B^3 \alpha_{wB0}}{360} + \frac{7L_B^3 \alpha_{wB1}}{360} - \frac{\beta_{wB0} L_B}{3} - \frac{\beta_{wB1} L_B}{6} \right) \\ + k_a \cos \theta \left(\sum_{m=0}^{\infty} C_{uB}^m - \frac{\beta_{uB0} L_B}{3} - \frac{\beta_{uB1} L_B}{6} \right) \end{aligned} \quad (35)$$

$$\begin{aligned} E_B S_B \beta_{uB0} - (k_a \cos^2 \theta + k \sin^2 \theta) \left(\sum_{m=0}^{\infty} C_{uB}^m - \frac{\beta_{uB0} L_B}{3} - \frac{\beta_{uB1} L_B}{6} \right) = \\ -k \sin^3 \theta \left[\sum_{m=0}^{\infty} (-1)^m C_{wA}^m - \frac{7L_A^3 \alpha_{wA0}}{360} - \frac{8L_A^3 \alpha_{wA1}}{360} + \frac{\beta_{wA0} L_A}{6} + \frac{\beta_{wA1} L_A}{3} \right] \\ -k_a \cos^3 \theta \left[\sum_{m=0}^{\infty} (-1)^m C_{uA}^m + \frac{\beta_{uA0} L_A}{6} + \frac{\beta_{uA1} L_A}{3} \right] \end{aligned} \quad (36)$$

From Eqs.(25)-(36) the unknown boundary parameters can be found as

$$\bar{\alpha} = \begin{Bmatrix} \bar{\alpha}_{wA} \\ \bar{\alpha}_{wB} \\ \bar{\alpha}_{uA} \\ \bar{\alpha}_{uB} \end{Bmatrix} = \sum_{m=0}^{\infty} \begin{Bmatrix} \widetilde{H}_1 Q_1^m C_{wA}^m + \widetilde{H}_1 Q_2^m C_{wA}^m + \widetilde{H}_1 Q_3^m C_{uA}^m + \widetilde{H}_1 Q_4^m C_{uB}^m \\ \widetilde{H}_2 Q_1^m C_{wA}^m + \widetilde{H}_2 Q_2^m C_{wA}^m + \widetilde{H}_2 Q_3^m C_{uA}^m + \widetilde{H}_2 Q_4^m C_{uB}^m \\ \widetilde{H}_3 Q_1^m C_{wA}^m + \widetilde{H}_3 Q_2^m C_{wA}^m + \widetilde{H}_3 Q_3^m C_{uA}^m + \widetilde{H}_3 Q_4^m C_{uB}^m \\ \widetilde{H}_4 Q_1^m C_{wA}^m + \widetilde{H}_4 Q_2^m C_{wA}^m + \widetilde{H}_4 Q_3^m C_{uA}^m + \widetilde{H}_4 Q_4^m C_{uB}^m \end{Bmatrix} \quad (37)$$

where

$$\widetilde{H} = \begin{Bmatrix} \widetilde{H}_1 \\ \widetilde{H}_2 \\ \widetilde{H}_3 \\ \widetilde{H}_4 \end{Bmatrix} = H^{-1} \quad (38)$$

with

$$H = \begin{bmatrix} H_{11} & H_{12} & 0 & H_{14} \\ H_{21} & H_{22} & H_{23} & 0 \\ 0 & H_{32} & H_{33} & H_{34} \\ H_{41} & 0 & H_{43} & H_{44} \end{bmatrix} \quad (39)$$

The detailed expressions for H_{ij} and Q_i ($i, j = 1, 2, 3, 4$) are given in Appendix.

Making the use of Eqs.(21)-(25) and (37), the flexural and axial displacements can finally be written as

$$w_A(x) = \sum_{m=0}^{\infty} C_{wA}^m \left[\cos(\lambda_{Am} x) + \zeta_{wA}(x)^T \widetilde{H}_1 Q_1^m \right] + \sum_{m=0}^{\infty} C_{wB}^m \zeta_{wA}(x)^T \widetilde{H}_1 Q_2^m + \sum_{m=0}^{\infty} C_{uB}^m \zeta_{wA}(x)^T \widetilde{H}_1 Q_4^m \quad (40)$$

$$w_B(x) = \sum_{m=0}^{\infty} C_{wB}^m \left[\cos(\lambda_{Bm} x) + \zeta_{wB}(x)^T \widetilde{H}_2 Q_2^m \right] + \sum_{m=0}^{\infty} C_{wA}^m \zeta_{wB}(x)^T \widetilde{H}_2 Q_1^m + \sum_{m=0}^{\infty} C_{uA}^m \zeta_{wB}(x)^T \widetilde{H}_2 Q_3^m \quad (41)$$

$$u_A(x) = \sum_{m=0}^{\infty} C_{uA}^m \left[\cos(\lambda_{Am}x) + \zeta_{uA}(x)^T \widetilde{\mathbf{H}}_3 \mathbf{Q}_3^m \right] + \sum_{m=0}^{\infty} C_{wB}^m \zeta_{uA}(x)^T \widetilde{\mathbf{H}}_3 \mathbf{Q}_2^m + \sum_{m=0}^{\infty} C_{uB}^m \zeta_{uA}(x)^T \widetilde{\mathbf{H}}_3 \mathbf{Q}_4^m \quad (42)$$

and

$$u_B(x) = \sum_{m=0}^{\infty} C_{uB}^m \left[\cos(\lambda_{Bm}x) + \zeta_{uB}(x)^T \widetilde{\mathbf{H}}_4 \mathbf{Q}_4^m \right] + \sum_{m=0}^{\infty} C_{wA}^m \zeta_{uB}(x)^T \widetilde{\mathbf{H}}_4 \mathbf{Q}_1^m + \sum_{m=0}^{\infty} C_{uA}^m \zeta_{uB}(x)^T \widetilde{\mathbf{H}}_4 \mathbf{Q}_3^m \quad (43)$$

Substituting Eqs.(40)-(43) into Eqs.(1)-(2) and following the Galerkin's procedure, one will be able to obtain

$$\begin{bmatrix} \mathbf{K}_{11} & \mathbf{K}_{12} & \mathbf{0} & \mathbf{K}_{14} \\ \mathbf{K}_{21} & \mathbf{K}_{22} & \mathbf{K}_{23} & \mathbf{0} \\ \mathbf{0} & \mathbf{K}_{32} & \mathbf{K}_{33} & \mathbf{K}_{34} \\ \mathbf{K}_{41} & \mathbf{0} & \mathbf{K}_{43} & \mathbf{K}_{44} \end{bmatrix} \begin{Bmatrix} C_{wA} \\ C_{wB} \\ C_{uA} \\ C_{uB} \end{Bmatrix} - \omega^2 \begin{bmatrix} \mathbf{M}_{11} & \mathbf{M}_{12} & \mathbf{0} & \mathbf{M}_{14} \\ \mathbf{M}_{21} & \mathbf{M}_{22} & \mathbf{M}_{23} & \mathbf{0} \\ \mathbf{0} & \mathbf{M}_{32} & \mathbf{M}_{33} & \mathbf{M}_{34} \\ \mathbf{M}_{41} & \mathbf{0} & \mathbf{M}_{43} & \mathbf{M}_{44} \end{bmatrix} \begin{Bmatrix} C_{wA} \\ C_{wB} \\ C_{uA} \\ C_{uB} \end{Bmatrix} = \begin{Bmatrix} \mathbf{F}_{wA} \\ \mathbf{F}_{wB} \\ \mathbf{F}_{uA} \\ \mathbf{F}_{uB} \end{Bmatrix} \quad (44)$$

where

$$\mathbf{C}_{p(r)q(r)} = \left\{ C_{p(r)q(r)}^0, C_{p(r)q(r)}^1, C_{p(r)q(r)}^2, \dots \right\}^T \quad (45)$$

$$\mathbf{F}_{p(r)q(r)} = \left\{ F_{p(r)q(r)}^0, F_{p(r)q(r)}^1, F_{p(r)q(r)}^2, \dots \right\}^T \quad (46)$$

$$\begin{aligned} K_{rs,mm'} &= \delta_{rs} (1 - \delta_{0m}) (1 - \delta_{0m'}) \left(\delta_{mm'} + S_{rs,m'm}^{p(r)q(r)} \right) \lambda_{q(r)m'}^4 \\ &\quad + \delta_{rs} \delta_{0m'} \delta_{m0} S_{rs,mm'}^{p(r)q(r)} + (1 - \delta_{rs}) \delta_{m0} S_{rs,mm'}^{p(r)q(r)} \end{aligned} \quad (47)$$

$$\begin{aligned} M_{rs,mm'} &= \delta_{rs} (1 - \delta_{0m}) (1 - \delta_{0m'}) \left(\delta_{mm'} + S_{rs,mm'}^{p(r)q(r)} + S_{rs,m'm}^{p(r)q(r)} + Z_{rs,mm'}^{p(r)q(r)} \right) \\ &\quad + \delta_{rs} \delta_{0m'} \delta_{m0} \left(2 + Z_{rs,mm'}^{p(r)q(r)} \right) + \delta_{rs} \delta_{0m'} \left(S_{rs,mm'}^{p(r)q(r)} + Z_{rs,mm'}^{p(r)q(r)} \right) \\ &\quad + \delta_{rs} \delta_{m0} \left(S_{rs,m'm}^{p(r)q(r)} + Z_{rs,mm'}^{p(r)q(r)} \right) + (1 - \delta_{rs}) (1 - \delta_{0m}) \left(S_{rs,mm'}^{p(r)q(r)} + Z_{rs,mm'}^{p(r)q(r)} \right) \\ &\quad + (1 - \delta_{rs}) \delta_{m0} Z_{rs,mm'}^{p(r)q(r)} \quad \text{for } m, m' = 0, 1, 2, 3, \dots \end{aligned} \quad (48)$$

and

$$F_{p(r)q(r),m} = \frac{2}{L_{q(r)}} \int_0^{L_{q(r)}} \left[\cos(\lambda_{q(r)m}x) + \zeta_{p(r)}(x)^T \widetilde{\mathbf{H}}_r \mathbf{Q}_{rm} \right] f_{p(r)q(r)}(x) dx \quad (49)$$

The matrices, $\mathbf{S}$ and $\mathbf{Z}$, in Eqs.(47) and (48) are also defined in Appendix.

Once the coefficients in Fourier expansions are solved from Eq.(44), the flexural and longitudinal displacements on each beam can be directly determined using Eqs.(40)-(43).

In above derivations, each beam is assumed to be uniform; that is, the cross-sectional areas, mass densities and bending rigidities are all constant. However, it should be pointed out that this condition is not necessary for the current solution method since the displacement expressions given in Eqs.(3) and (4) are equally valid for a non-uniform beam. When a non-uniform beam is involved, only the governing equations and boundary/coupling conditions related to that particular beam need to be modified accordingly; all solution algorithms and procedures will be essentially the same except that the integrands will now explicitly contain these material or geometric variables. Thus, the current method can be directly extended to non-uniform beams.

III. RESULTS AND DISCUSSIONS

As an example, consider two beams connected together at a 45° angle, as shown in Fig.2. The left end of the horizontal beam (beam A) is fixed onto the ground, and the second beam is simply supported at the right end. The stiffness values for the coupling springs are also shown in Fig.2. The remaining properties for these two beams are summarized in Table 1.

For model validation, this problem is solved using both the current method and the commercial finite element code, ANSYS^[21]. The FEA model consists of 250 beam elements of equal size, 0.01 m,

Table 1. Beam parameters

Properties	Beam 1	Beam 2
Length (m)	1.0	1.5
Young's Modulus (N/m ²)	$2.07 \times 10^{+11}$	$2.07 \times 10^{+11}$
Cross-Section Area (m ²)	5.0×10^{-5}	1.5×10^{-5}
Moment of Inertia (Kg.m ⁴)	1.0×10^{-10}	5.0×10^{-11}
Density (Kg/m ³)	$7.8 \times 10^{+3}$	$7.8 \times 10^{+3}$

which is considered adequately fine to capture the spatial variations of the displacements along each beam. The first 10 natural frequencies for this two-beam system are compared in Table 2. An excellent agreement is observed between these two sets of solutions with a maximum difference less than 0.5%.

Table 2. A comparison of the first 10 natural frequencies

Modes	Current	FEA	Difference (%)
1	8.836	8.829	0.069
2	13.398	13.344	0.408
3	28.011	27.932	0.282
4	35.402	35.422	0.057
5	61.775	61.709	0.108
6	80.624	80.663	0.049
7	107.954	107.87	0.078
8	148.291	148.31	0.013
9	164.309	164.21	0.060
10	223.558	223.23	0.147

Next example involves two identical beams rigidly connected at right angles, as sketched in Fig.3. The relevant properties for the beams are as follows: Young's modulus $E = 2.07 \times 10^{+11}$ Pa, Poisson ratio $\nu = 0.3$, mass density $\rho = 7800$ kg/m³, bending moment $I = 2.3368 \times 10^{-7}$ m⁴, cross-section area $S = 8.5 \times 10^{-4}$ m² and length $L = 2.98$ m.

Vibration tests were performed on this structure with a complete free boundary condition. To simulate the free boundary condition, the rigidly coupled beams are placed on several soft foam blocks. A dozen of accelerators are 'randomly' placed on the structure, as identified in Fig.3. The shaker is attached to beam A, via an impedance head, at a distance 1.05 m from the left end, as marked by an arrow in Fig.3. The beams have been carefully connected together to maintain a symmetry about the plane defined by the center lines of the beams. In order to avoid exciting the out-of-plane modes or causing warping motions, the shaker head is attached perpendicularly and aligned to the center line of beam A so that the excitation force can be applied in the horizontal

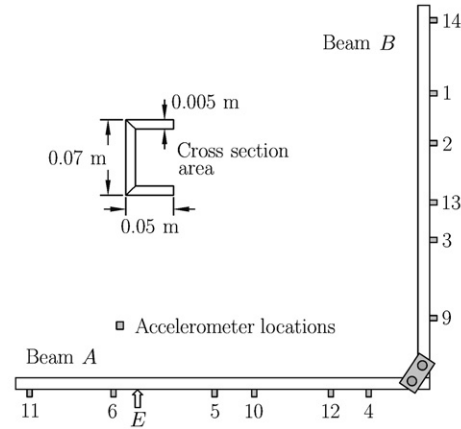


Fig. 3 Beam dimensions, and the sensor and shaker (labeled as E) locations.

direction. Thus, the resulting vibrations will be predominantly in the plane of symmetry. A broad-band random signal is fed to the shaker so that many modes can be simultaneously excited and measured. Plotted in Fig.4 are the measured and predicted frequency response functions (FRF) for two selected sensor locations, one on each beam. Because of the rigid joints (or strong coupling), most modes are global as manifested in the fact that the corresponding peaks simultaneously show on both response curves. It is seen that the predicted and measured responses compare reasonably well except that: 1) the measured FRF data seem to be severely contaminated at frequencies below 70 Hz; and 2) there is a significant frequency-shift (almost 10%) between measured and predicted peaks near 250 Hz. The

contamination problem is believed due to the poor response characteristic of the shaker (or the poor signal-to-noise ratios) at the low frequencies. In comparison, the cause(s) for the peak-shaft cannot be easily determined. The possible explanations may include: 1) the non-uniformity of the actual beams, 2) the deviation of the joint from being perfectly rigid, 3) the effective lengths of the beams vary with frequency (refer to Fig.3 for the joint configuration), and 4) the cross-coupling between the in-plane and out-of-plane modes. It may also be speculated that the 'spurious' peak near 375 Hz with the measured FRF's is related to the out-of-plane vibration. It must be pointed out, however, that the frequency mismatches between the measured and predicted peaks should not be viewed as an indication of the possible problems with the current formulations. In essence, the modal results given in Table 2 are considered as accurate enough from the viewpoint of model verification.

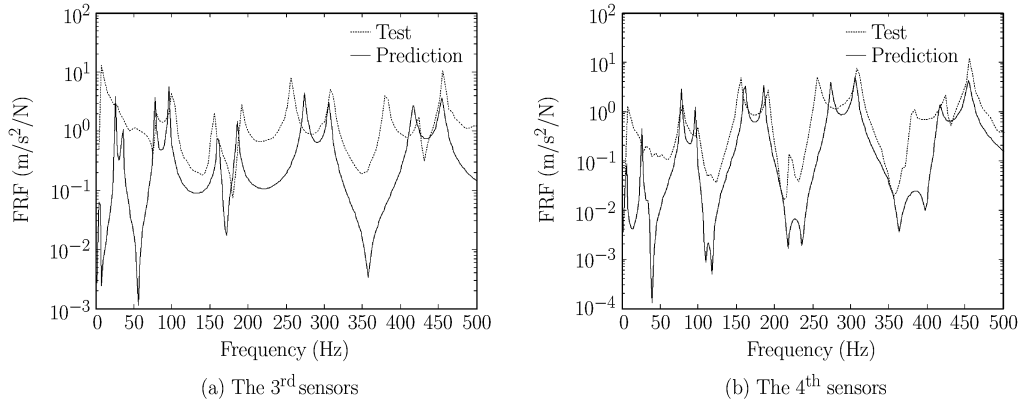


Fig. 4. Vibrations at: (a) the 3rd and (b) 4th sensors, which are located at 1.0 m and 0.5 m from the junction, respectively.

The natural frequencies of the coupled beam system can be simply identified from the peaks in the FRF curves. For example, the modes shapes of two of the identified modes are plotted in Figs.5 and 6. The data seem to support above speculations regarding the non-uniformity of the beams used in the test and the frequency-dependence of their effective lengths.

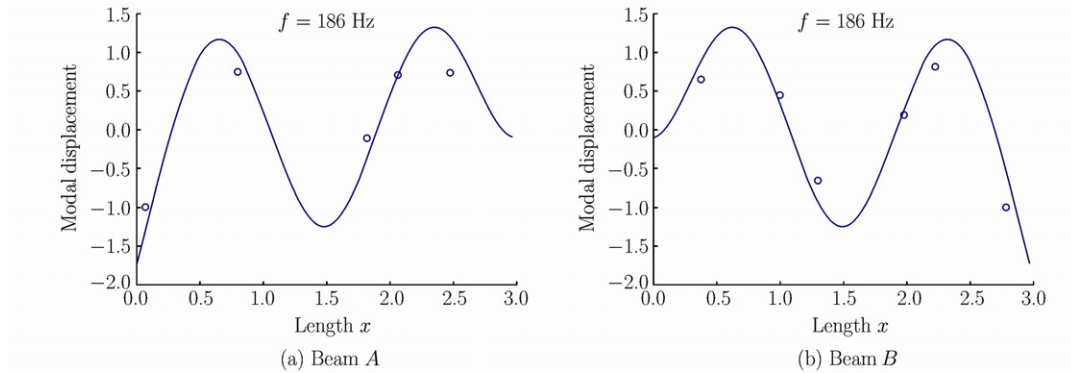


Fig. 5. The mode shapes for natural frequency 186 Hz.

IV. CONCLUSIONS

A general analysis method has been developed for determining the vibrations of two beams elastically coupled together at an arbitrary angle. The coupling between the beams is generically represented by a set of springs with arbitrary stiffnesses, and the coupling conditions based on the force equilibriums are exactly enforced at the junction without using any assumptions or approximations. Thus, vibrations of

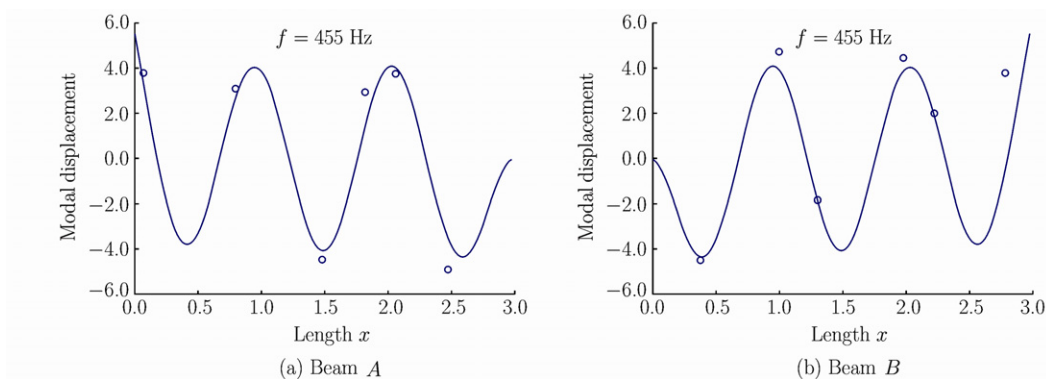


Fig. 6. The mode shapes for natural frequency 455 Hz.

the coupled beam system can be accurately determined regardless of whether the coupling is weak or strong. The cross-coupling between flexural and longitudinal waves at the junction is also considered in a general manner. Since the beams are elastically restrained at each end, the current method can be universally applied to various boundary conditions with no need of modifying the basis functions as often required in other methods. Finally, it should be noted that, although this investigation is only focused on the coupling of two uniform beams, there is no inherent difficulty in extending the current method to a system consisting of any number of uniform or non-uniform beams.

References

- [1] Hambric, S.A., Power flow and mechanical intensity calculations in structural finite element analysis. *Journal of Vibration and Acoustics*, 1990, 112: 542-549.
- [2] Gavric, L. and Pavic, G., A finite element method for computation of structural intensity by the normal mode approach. *Journal of Sound and Vibration*, 1993, 164: 29-43.
- [3] Szwer, R.P., Burroughs, C.B., Hambric, S.A. and McDevitt, T.E., Power flow in coupled bending and longitudinal waves in beams. *Journal of the Acoustical Society of America*, 2000, 107: 3186-3195.
- [4] Mace, B.R. and Shorter, P., Energy flow models from finite element analysis. *Journal of Sound and Vibration*, 2000, 233: 369-389.
- [5] Rabbio, G., Bernhard, R.J. and Milner, F.A., Definition of a high-frequency threshold for plates and acoustical spaces. *Journal of Sound and Vibration*, 2004, 277: 647-667.
- [6] Bercin, A.N. and Langley, R.S., Application of the dynamic stiffness technique to the inplane vibrations of plate structures. *Computers and Structures*, 1996, 59: 869-875.
- [7] Langley, R.S., Analysis of power flow in beams and frameworks using the direct-dynamic stiffness method. *Journal of Sound and Vibration*, 1990, 136: 439-452.
- [8] Park, D.H., Hong, S.Y., Kil, F.G. and Jeon, J.J., Power flow models and analysis of in-plane waves in finite coupled thin plates. *Journal of Sound and Vibration*, 2001, 244: 651-668.
- [9] Doyle, J.F., *Wave Propagation in Structures*. Springer, Berlin, 1989.
- [10] Ahmida, K.M. and Arruda, J.R.F., Spectral element-based prediction of active power flow in Timoshenko beams. *International Journal of Solids and Structures*, 2001, 38: 1669-1679.
- [11] Igawa, H., Komatsu, K., Yamaguchi, I. and Kasai, T., Wave propagation analysis of frame structures using the spectral element method. *Journal of Sound and Vibration*, 2004, 277: 1071-1081.
- [12] Keane, A.J. and Price, W.G., A note on the power flowing between two conservatively coupled multi-modal sub-system. *Journal of Sound and Vibration*, 1991, 144: 185-196.
- [13] Keane, A.J., Energy flows between arbitrary configurations of conservatively coupled multi-modal elastic subsystems. *Proceedings of the Royal Society of London A*, 1992, 436: 537-568.
- [14] Beshara, M. and Keane, A.J., Vibrational power flows in beam networks with compliant and dissipative joints. *Journal of Sound and Vibration*, 1997, 203: 321-339.
- [15] Shankar, K. and Keane, A.J., Power flow predictions in a structure of rigidly joined beams using receptance theory. *Journal of Sound and Vibration*, 1995, 180: 867-890.
- [16] Farag, N.H. and Pan, J., Dynamic response and power flow in three-dimensional coupled beam structures. I. Analytical modeling. *Journal of the Acoustical Society of America*, 1997, 102: 315-325.
- [17] Keane, A.J., Energy flows between arbitrary configurations of conservatively coupled multi-modal elastic subsystems. *Proceedings of the Royal Society of London A*, 1992, 436: 537-568.

- [18] Li,W.L., Bonilha,M.W. and Xiao,J., Vibrations and power flows in a coupled beam system. *Journal of Vibration and Acoustics*, 2007, 129: 617-622.
- [19] Li,W.L., Free vibration of beams with general boundary conditions. *Journal of Sound and Vibration*, 2000, 237: 709-725.
- [20] Li,W.L., Comparison of Fourier sine and cosine series expansions for beams with arbitrary boundary conditions. *Journal of Sound and Vibration*, 2002, 255: 185-194.
- [21] ANSYS Release 6.1. ANSYS Structural Analysis Guide. ANSYS Inc, 2002.

APPENDIX

Definitions and Expressions of $\mathbf{H}$, $\mathbf{Q}$, $\mathbf{S}$ and $\mathbf{Z}$ Vectors/Matrices

The sub-matrices of the $\mathbf{H}$ matrix in Eq.(39) are defined as follows:

$$\mathbf{H}_{11} = \begin{bmatrix} 8k_0L_A^3/360 + D_A & 7k_0L^3/360 & -k_0L_A/3 & -k_0L_A/6 \\ 7kL_A^3/360 & 8kL_A^3/360 + D_A & -kL_A/360 & -kL_A/3 \\ L_AD_A/3 & L_AD_A/6 & K_0 + D_A/L_A & -D_A/L_A \\ L_AD_A/6 & L_AD_A/3 & -D_A/L_A & K + D_A/L_A \end{bmatrix} \quad (50)$$

$$\mathbf{H}_{12} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 8kL_B^3 \cos \theta/360 & 7kL_B^3 \cos \theta/360 & -kL_B \cos \theta/3 & -kL_B \cos \theta/6 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -K & 0 \end{bmatrix} \quad (51)$$

$$\mathbf{H}_{14} = \begin{bmatrix} 0 & 0 \\ -kL_B \sin \theta/3 & -kL_B \sin \theta/6 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \quad (52)$$

$$\mathbf{H}_{21} = \begin{bmatrix} 7kL_A^3 \cos^3 \theta/360 & 8kL_A^3 \cos^3 \theta/360 & -kL_A \cos^3 \theta/6 & -kL_A \cos^3 \theta/3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -K \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad (53)$$

$$\mathbf{H}_{22} = \begin{bmatrix} 8\bar{k}L_B^3/360 + D_B & 7\bar{k}L_B^3/360 & -\bar{k}L_B/3 & -\bar{k}L_B/6 \\ 7k_1L_B^3/360 & 8k_1L_B^3/360 + D_B & -k_1L_B/6 & -k_1L_B/3 \\ L_B D_B/3 & L_B D_B/6 & K + D_B/L_B & -D_B/L_B \\ L_B D_B/6 & L_B D_B/3 & -D_B/L_B & K_1 + D_B/L_B \end{bmatrix} \quad (54)$$

$(\bar{k} = k \cos^2 \theta + k_a \sin^2 \theta)$

$$\mathbf{H}_{23} = \begin{bmatrix} k_a L_A \sin^3 \theta/6 & k_a L_A \sin^3 \theta/3 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \quad (55)$$

$$\mathbf{H}_{32} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 8k_a L_B^3 \sin \theta/360 & 7k_a L_B^3 \sin \theta/360 & -k_a L_B \sin \theta/3 & -k_a L_B \sin \theta/6 \end{bmatrix} \quad (56)$$

$$\mathbf{H}_{33} = \begin{bmatrix} k_{a0}L_a/3 + E_a S_a & k_{a0}L_a/6 \\ k_a L_a/6 & k_a L_a/3 + E_a S_a \end{bmatrix} \quad (57)$$

$$\mathbf{H}_{34} = \begin{bmatrix} 0 & 0 \\ k_a L_b \cos \theta/3 & k_a L_b \cos \theta/6 \end{bmatrix} \quad (58)$$

$$\mathbf{H}_{41} = \begin{bmatrix} -7kL_A^3 \sin^3 \theta/360 & -8kL_A^3 \sin^3 \theta/360 & kL_A \sin^3 \theta/6 & kL_A \sin^3 \theta/3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad (59)$$

$$\mathbf{H}_{43} = \begin{bmatrix} k_a L_a \cos^3 \theta/6 & k_a L_a \cos^3 \theta/3 \\ 0 & 0 \end{bmatrix} \quad (60)$$

$$\mathbf{H}_{44} = \begin{bmatrix} \bar{k}_a L_b/3 + E_b S_b & \bar{k}_a L_b/6 \\ k_{a1} L_a/6 & k_{a1} L_b/3 + E_b S_b \end{bmatrix} \quad (\bar{k}_a = k_a \cos^2 \theta + k \sin^2 \theta) \quad (61)$$

The $\mathbf{Q}$ -vectors in Eq.(37) are defined as

$$\mathbf{Q}_1^m = \{ -k_0 \ (-1)^m k \ -D_A \lambda_{Am}^2 \ (-1)^m D_A \lambda_{Am}^2 \ (-1)^m k \cos^3 \theta \ 0 \ 0 \ 0 \ 0 \ 0 \ (-1)^{m+1} k \sin^3 \theta \ 0 \}^T \quad (62)$$

$$\mathbf{Q}_2^m = \{ 0 \ -k \cos \theta \ 0 \ 0 \ -\bar{k} \ (-1)^m k_1 \ -D_B \lambda_{Bm}^2 \ (-1)^m D_B \lambda_{Bm}^2 \ 0 \ -k_a \sin \theta \ 0 \ 0 \}^T \quad (63)$$

$$\mathbf{Q}_3^m = \{ 0 \ 0 \ 0 \ 0 \ (-1)^{m+1} k_a \sin^3 \theta \ 0 \ 0 \ 0 \ k_{a0} \ (-1)^{m+1} k_a \ (-1)^{m+1} k_a \cos^3 \theta \ 0 \}^T \quad (64)$$

and

$$\mathbf{Q}_4^m = \{ 0 \ -k \sin \theta \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ k_a \cos \theta \ \bar{k}_a \ (-1)^{m+1} k_{a1} \}^T \quad (65)$$

The matrices and vectors used in Eqs.(47) and (48) are defined as follows:

$$S_{rs,mm'}^{p(r)q(r)} = \mathbf{P}_m^{pq} \widetilde{\mathbf{H}}_r \mathbf{Q}_{sm'} \quad (p = A, B; q = w, u; r, s = 1, 2, 3, 4) \quad (66)$$

$$Z_{rs,mm'}^{p(r)q(r)} = \mathbf{Q}_{rm}^T \widetilde{\mathbf{H}}_r^T \mathbf{\Xi}_q^p \widetilde{\mathbf{H}}_r \mathbf{Q}_{sm'} \quad (67)$$

where

$$p(r) = \begin{cases} w & \text{for } r = 1, 2 \\ u & \text{for } r = 3, 4 \end{cases} \quad (68)$$

$$q(r) = \begin{cases} A & \text{for } r = 1, 3 \\ B & \text{for } r = 2, 4 \end{cases} \quad (69)$$

$$\mathbf{P}_m^{wq} = \begin{cases} \frac{2}{L_q} \{ 1/\lambda_{qm}^4 \ (-1)^{m+1}/\lambda_{qm}^4 \ -1/\lambda_{qm}^2 \ (-1)^m/\lambda_{qm}^2 \} & (m \neq 0) \\ \frac{2}{L_q} \{-1 \ 1 \ 0 \ 0\} & (m = 0) \end{cases} \quad (70)$$

$$\mathbf{P}_m^{uq} = \begin{cases} \frac{2}{L_q} \{ -1/\lambda_{qm}^2 \ (-1)^m/\lambda_{qm}^2 \} & (m \neq 0) \\ \frac{2}{L_q} \{-1 \ 1\} & (m = 0) \end{cases} \quad (71)$$

$$\mathbf{\Xi}_q^w = \frac{2}{L_q} \int_0^{L_q} \boldsymbol{\zeta}_{wq}(x)^T \boldsymbol{\zeta}_{wq}(x) dx = \begin{bmatrix} 2L_q^6/4725 & & & \\ 127L_q^6/302400 & 2L_q^6/4725 & \text{sym.} & \\ -4L_q^4/945 & -31L_q^4/7560 & 2L_q^2/45 & \\ -31L_q^4/7560 & -4L_q^4/945 & 7L_q^2/180 & 2L_q^2/45 \end{bmatrix} \quad (72)$$

$$\mathbf{\Xi}_q^u = \frac{2}{L_q} \int_0^{L_q} \boldsymbol{\zeta}_{uq}(x)^T \boldsymbol{\zeta}_{uq}(x) dx = \begin{bmatrix} 2L_q^2/45 & 7L_q^2/180 \\ 7L_q^2/180 & 2L_q^2/45 \end{bmatrix} \quad (73)$$

Additionally, the following identities have been used in deriving Eq.(44):

$$S_{11,0m'}^{wA} + \lambda_{Am'}^4 S_{11,m'0}^{wA} \equiv 0 \quad (74)$$

$$S_{22,0m'}^{wB} + \lambda_{Bm'}^4 S_{22,m'0}^{wB} \equiv 0 \quad (75)$$

$$S_{33,0m'}^{uA} - \lambda_{Am'}^2 S_{33,m'0}^{uA} \equiv 0 \quad (76)$$

and

$$S_{44,0m'}^{uB} - \lambda_{Bm'}^2 S_{44,m'0}^{uB} \equiv 0 \quad (77)$$